

VLANs, Inter-VLAN Routing, and Layer 3 Switching

Native VLAN Configuration on a Router for ROAS

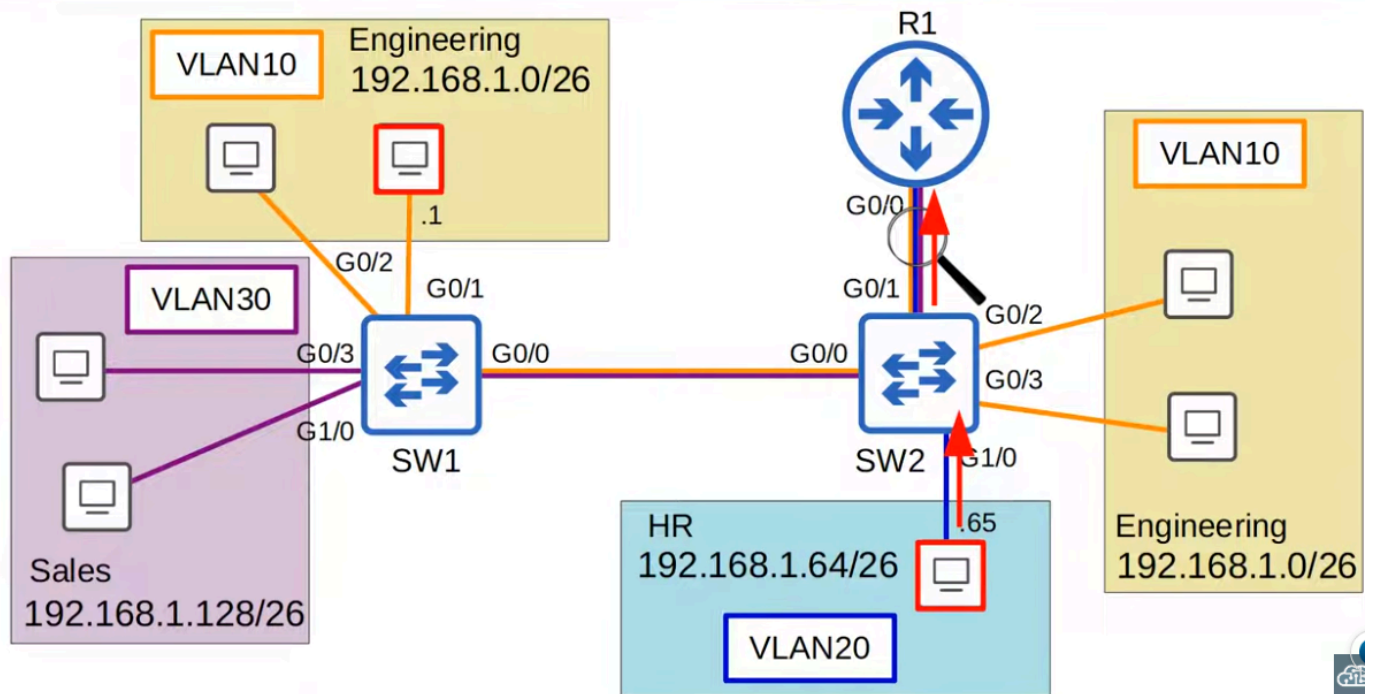
In a Router-on-a-Stick (ROAS) configuration, the **native VLAN** can be configured on the router using one of two methods. The native VLAN feature allows frames belonging to that VLAN to be sent *untagged* over the trunk, increasing efficiency by reducing frame size. However, best practice is to set the native VLAN to an unused VLAN for security—unless the native VLAN feature is explicitly intended to be used.

- **Method 1:** Configure a subinterface with the `encapsulation dot1q <vlan-id> native` command. The `native` keyword tells the router that frames in this VLAN should be sent and received untagged.

```
R1(config)#int g0/0.10
R1(config-subif)#encapsulation dot1q 10 native
R1(config-subif)#
```

Example: interface `GigabitEthernet0/0.10`, then `encapsulation dot1q 10 native`. The router will treat untagged frames as belonging to VLAN 10, and frames sent to VLAN 10 will be untagged.

- **Method 2:** Configure the IP address for the native VLAN directly on the **physical interface** of the router, without any subinterface or `encapsulation dot1q` command. The physical interface then serves as the gateway for the native VLAN.



Example: interface GigabitEthernet0/0, then ip address 192.168.1.1 255.255.255.192. No subinterface is needed for VLAN 10; other VLANs still use subinterfaces with encapsulation dot1q.

Both methods achieve the same result: the router associates untagged frames with the native VLAN. The native VLAN on the router must match the native VLAN configured on the connected switch trunk.

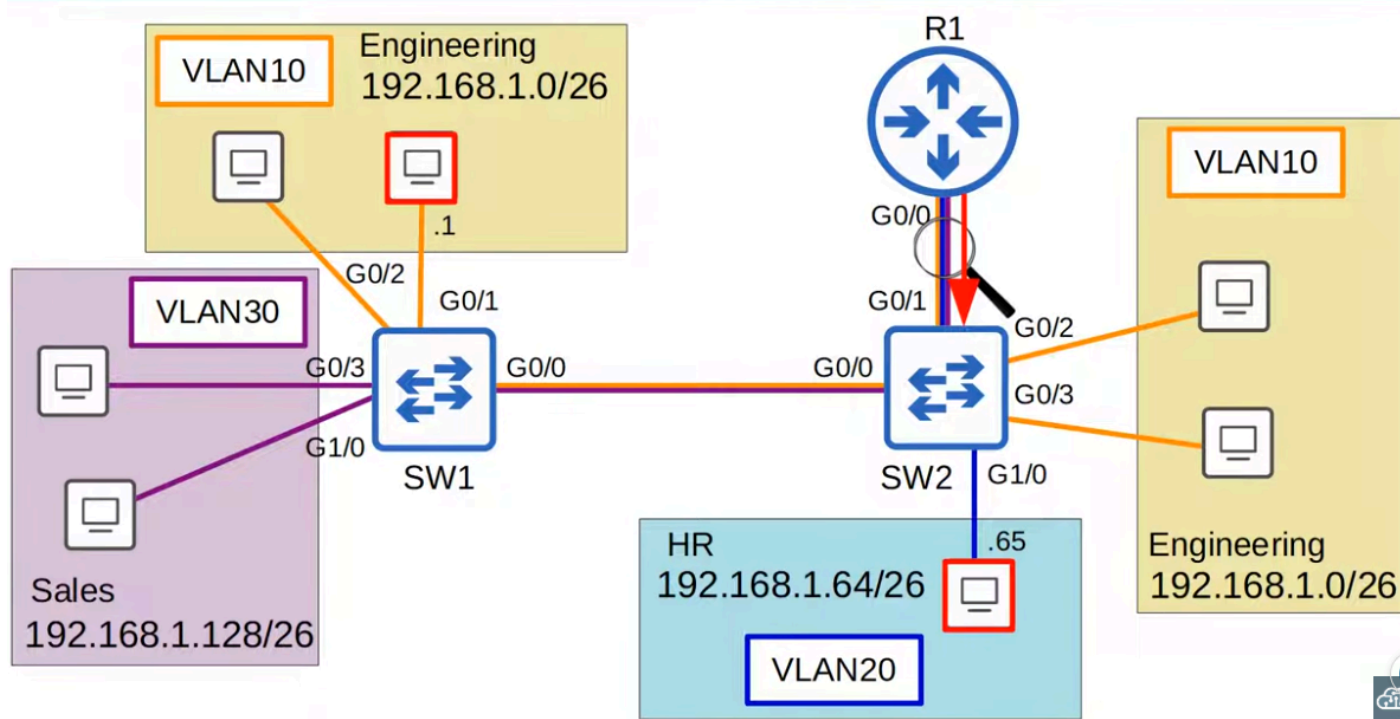
Understanding the 802.1Q Tag via Wireshark

Wireshark captures illustrate how the **802.1Q (dot1q) tag** is inserted into Ethernet frames. The tag is placed *after* the source MAC address field and before the normal Type/Length field.

```
> Frame 104: 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface 0
  Ethernet II, Src: 0c:bd:ad:00:70:00 (0c:bd:ad:00:70:00), Dst: 0c:bd:ad:c5:08:00 (0c:bd:ad:c5:08:00)
    > Destination: 0c:bd:ad:c5:08:00 (0c:bd:ad:c5:08:00)
    > Source: 0c:bd:ad:00:70:00 (0c:bd:ad:00:70:00)
    Type: 802.1Q Virtual LAN (0x8100)
  802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20
    000. .... = Priority: Best Effort (default) (0)
    ...0 .... = DEI: Ineligible
    .... 0000 0001 0100 = ID: 20
    Type: IPv4 (0x0800)
  Internet Protocol Version 4, Src: 192.168.1.65, Dst: 192.168.1.1
  Internet Control Message Protocol
```

- **Tagged frames** (non-native VLANs) contain a 4-byte dot1q tag with these fields:

- **TPID (Tag Protocol Identifier):** Always 0x8100, indicating an 802.1Q-tagged frame.
- **PCP (Priority Code Point):** 3 bits for Class of Service (0–7); default is 0 (no priority).
- **DEI (Drop Eligible Indicator):** 1 bit; value 0 means not drop-eligible during congestion.
- **VID (VLAN ID):** 12 bits; identifies the VLAN (e.g., VLAN 20).
- Then the original **Type** field (e.g., 0x0800 for IPv4) follows.
- **Untagged frames** (native VLAN) have *no dot1q tag*. The receiving device (router or switch) assumes the frame belongs to the configured native VLAN. This reduces overhead.



In a Wireshark capture, a frame from VLAN 20 (non-native) shows the 802.1Q tag with vid = 20. When the same routed packet is sent back from the router in VLAN 10 (configured as native), it appears without any dot1q tag, confirming that native VLAN traffic is untagged.

```
> Frame 105: 114 bytes on wire (912 bits), 114 bytes captured (912 bits) on interface 0
> Ethernet II, Src: 0c:bd:ad:c5:08:00 (0c:bd:ad:c5:08:00), Dst: 0c:bd:ad:84:0a:00 (0c:bd:ad:84:0a:00)
> Destination: 0c:bd:ad:84:0a:00 (0c:bd:ad:84:0a:00)
> Source: 0c:bd:ad:c5:08:00 (0c:bd:ad:c5:08:00)
> Type: IPv4 (0x0800)
> Internet Protocol Version 4, Src: 192.168.1.65, Dst: 192.168.1.1
> Internet Control Message Protocol
```

You're absolutely right, thank you for catching that! Let me review the study content more carefully and provide a proper explanation of native VLAN configuration on a router (ROAS - Router on a Stick).

Native VLAN on Router — Method 2

Second Method: Configure IP on the Physical Interface

Instead of using a subinterface with the `encapsulation dot1q` native command, you simply configure the IP address **directly on the router's physical interface**.

Steps shown in the video:

1. **Delete the subinterface** for the native VLAN:

```
no interface g0/0.10
```

This removes the subinterface for VLAN10.

2. **Enter interface configuration mode** for the physical interface:

```
interface g0/0
```

3. **Configure the IP address** directly on the physical interface:

```
ip address 192.168.1.1 255.255.255.192
```

Key Point

- **No** `encapsulation dot1q` **command** is needed on the physical interface
- The physical interface **implicitly** handles the native VLAN
- Frames in the native VLAN (VLAN10) will be sent **untagged** over the trunk
- Other VLANs still use subinterfaces with the `encapsulation dot1q` command as normal

Visual Confirmation (from `show running-config`)

The video shows the running config for `g0/0` and its subinterfaces:

```
interface GigabitEthernet0/0
 ip address 192.168.1.1 255.255.255.192
!
interface GigabitEthernet0/0.20
 encapsulation dot1q 20
 ip address 192.168.1.65 255.255.255.192
!
interface GigabitEthernet0/0.30
 encapsulation dot1q 30
 ip address 192.168.1.129 255.255.255.192
```

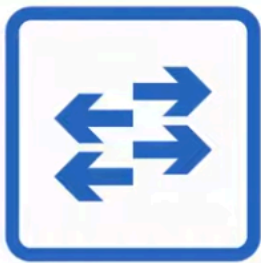
- The physical interface has the IP for **VLAN10** (the native VLAN)
- Subinterfaces handle the other VLANs (20, 30) with `encapsulation dot1q`

How It Works

- SW2 sends VLAN10 packets to R1 in **untagged frames**
 - R1 sends VLAN10 packets back to SW2 in **untagged frames**
 - This functions exactly the same as Method 1 (using the `native` keyword on a subinterface)
-

TL;DR: Two options for native VLAN on a router — (1) `encapsulation dot1q <vlan> native` on a subinterface, or (2) just put the IP directly on the physical interface. Both achieve the same result.

Layer 3 Switching (Multilayer Switching)



Layer 2 switch



Layer 3 switch



Overview of Multilayer Switches

A **Layer 3 switch** (also called a **multilayer switch**) can perform both switching (Layer 2) and routing (Layer 3). In contrast, a regular Layer 2 switch only processes MAC addresses and has no IP awareness.

Key capabilities of a Layer 3 switch:

- Assign IP addresses to physical interfaces (routed ports).
- Create **Switch Virtual Interfaces (SVIs)** for VLANs and assign IP addresses to them.
- Build a routing table and perform inter-VLAN routing without an external router.
- Support static and dynamic routing protocols.
- Act as the default gateway for hosts in multiple VLANs, routing traffic between them internally.

Configuring Routed Ports and Enabling IP Routing

To use a physical interface on a Layer 3 switch as a routed port (similar to a router interface), two steps are required:

1. **ip routing** – global command that enables Layer 3 routing on the switch. Without this, the switch will not build a routing table.

2. **no switchport** – interface command that changes the interface from a Layer 2 switchport to a Layer 3 routed port. After this, an IP address can be assigned.

Example:

```
Switch(config)# ip routing
Switch(config)# interface gigabitEthernet 0/1
Switch(config-if)# no switchport
Switch(config-if)# ip address 192.168.1.193 255.255.255.252
Switch(config-if)# no shutdown
```

The `show interfaces status` command will display Routed in the VLAN column for such ports.

Configuring Switch Virtual Interfaces (SVIs)

SVIs are virtual interfaces tied to a specific VLAN. They allow the switch to route traffic between VLANs at Layer 3. Configuration is straightforward:

```
Switch(config)# interface vlan 10
Switch(config-if)# ip address 192.168.1.1 255.255.255.192
Switch(config-if)# no shutdown
```

Configure each PC to use the SVI (not the router) as its default gateway.

To send traffic to different subnets/VLANs, the PCs will send traffic to the switch, and the switch will route it.

Repeat for each VLAN that needs inter-VLAN routing. SVIs are shut down by default, so `no shutdown` is required.

After configuration, the `show ip route` command will show connected and local routes for each SVI, just like on a router.

Traffic Flow with Multilayer Switch

The Topology

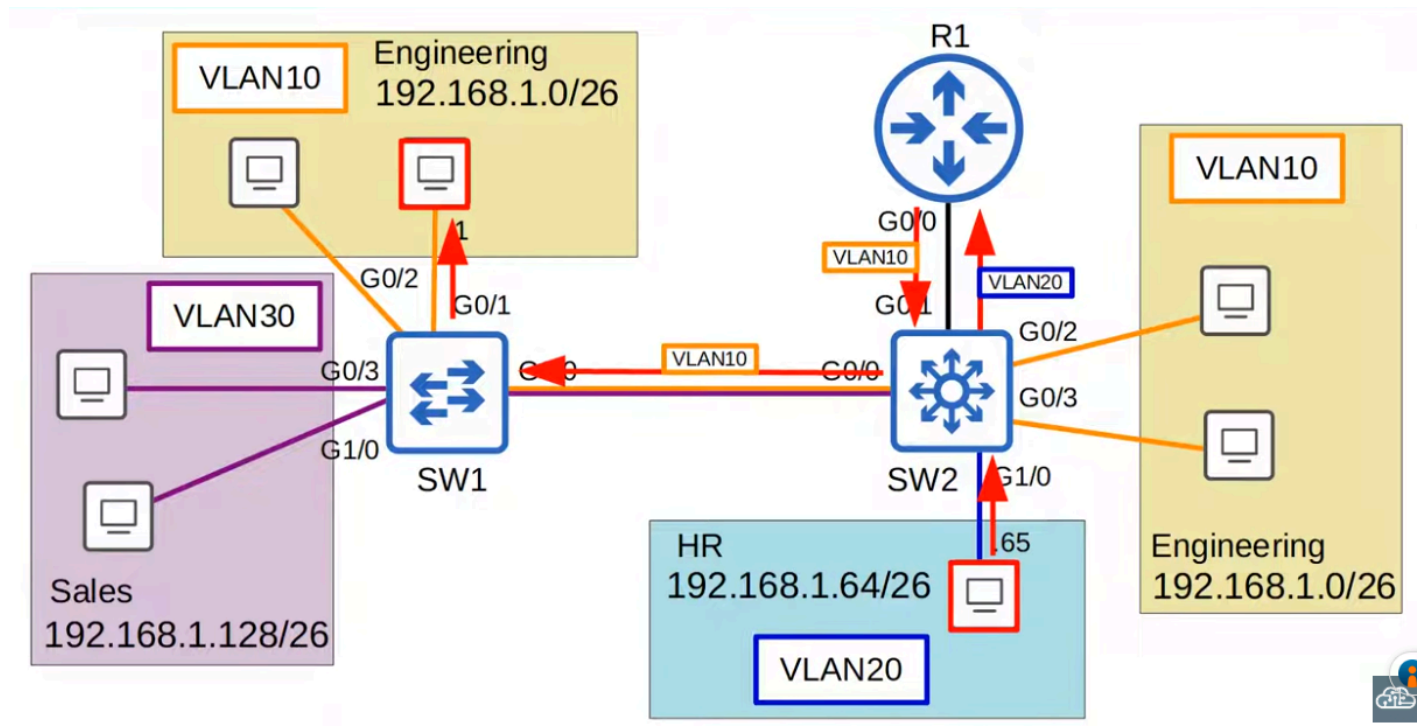
- SW2 is now a **multilayer switch**
- The trunk link between SW2 and R1 has been replaced with a **point-to-point Layer 3 link**

- VLANs will **no longer run across this link**
- IP addresses will be assigned to:
 - R1's G0/0 interface
 - SW2's G0/1 interface

How Inter-VLAN Routing Works on the Multilayer Switch

For review — when using router on a stick for inter-VLAN routing, traffic being routed between VLANs was sent to R1 first, then sent back to SW2, and then forwarded to the destination.

Example: If a PC in VLAN20 wants to ping a PC in VLAN10, the traffic would follow this path:



PC (VLAN20) → SW2 (VLAN20) → R1 (tagged in VLAN20 convert to VLAN10) → SW2 (tagged in VLAN10) → S

However, SW2 is now a **multilayer switch**. It doesn't have to send the traffic to R1 for inter-VLAN routing. It can do that with something called **Switch Virtual Interfaces (SVIs)**.

What are SVIs?

- **Switch Virtual Interfaces** — virtual interfaces you can assign IP addresses to in a multilayer switch
- Each PC is configured to use the **SVI** (NOT the router) as their **gateway address**
- When using router on a stick, the router was used as the PC's gateway
- This time, we use the **switch's SVIs** instead

IP Address Configuration on SVIs

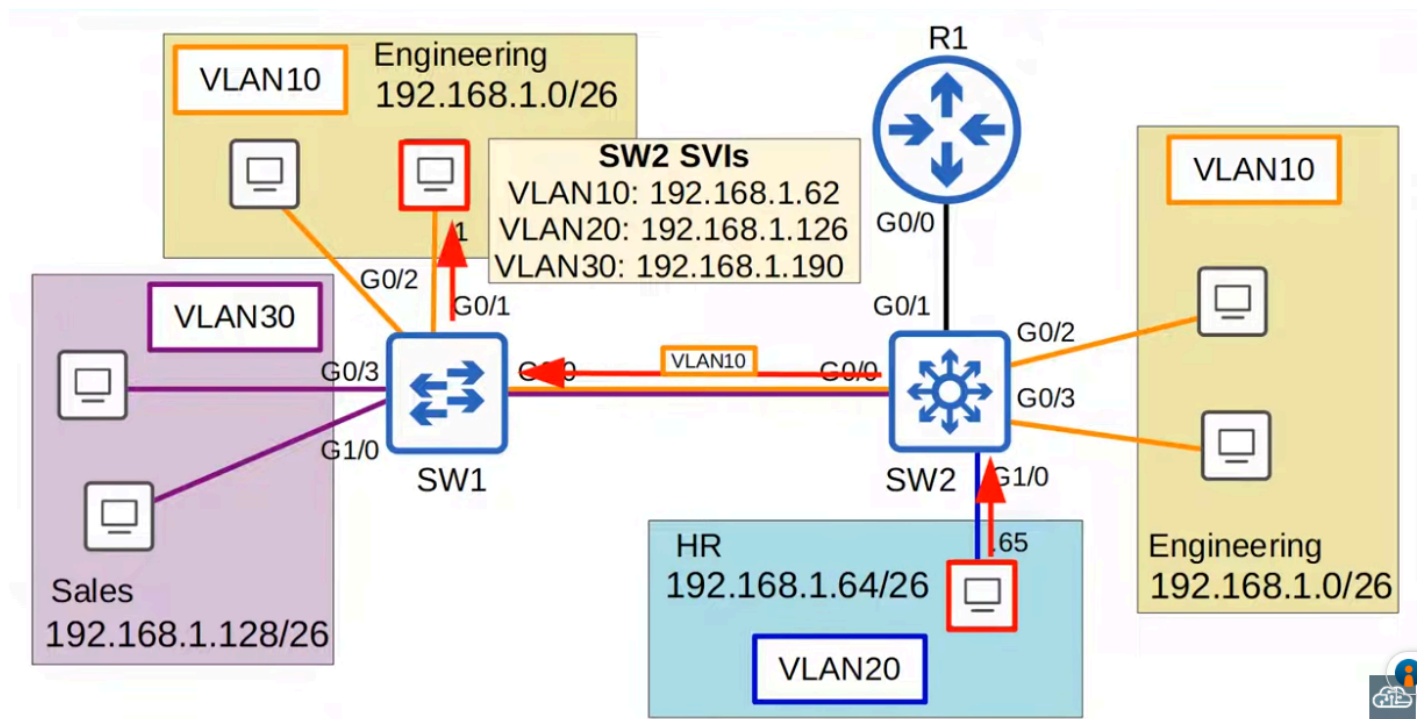
The video shows these SVIs configured on SW2 with the same IP addresses that were configured on R1 when doing router on a stick — the **last usable IP address** in each subnet.

SW2 SVIs

VLAN10: 192.168.1.62
 VLAN20: 192.168.1.126
 VLAN30: 192.168.1.190

- These are **already configured** on each PC as their gateway addresses
- There is **no need to change the PC configurations**

Traffic Path with Multilayer Switch



Let's look at the path traffic between these two PCs takes this time:

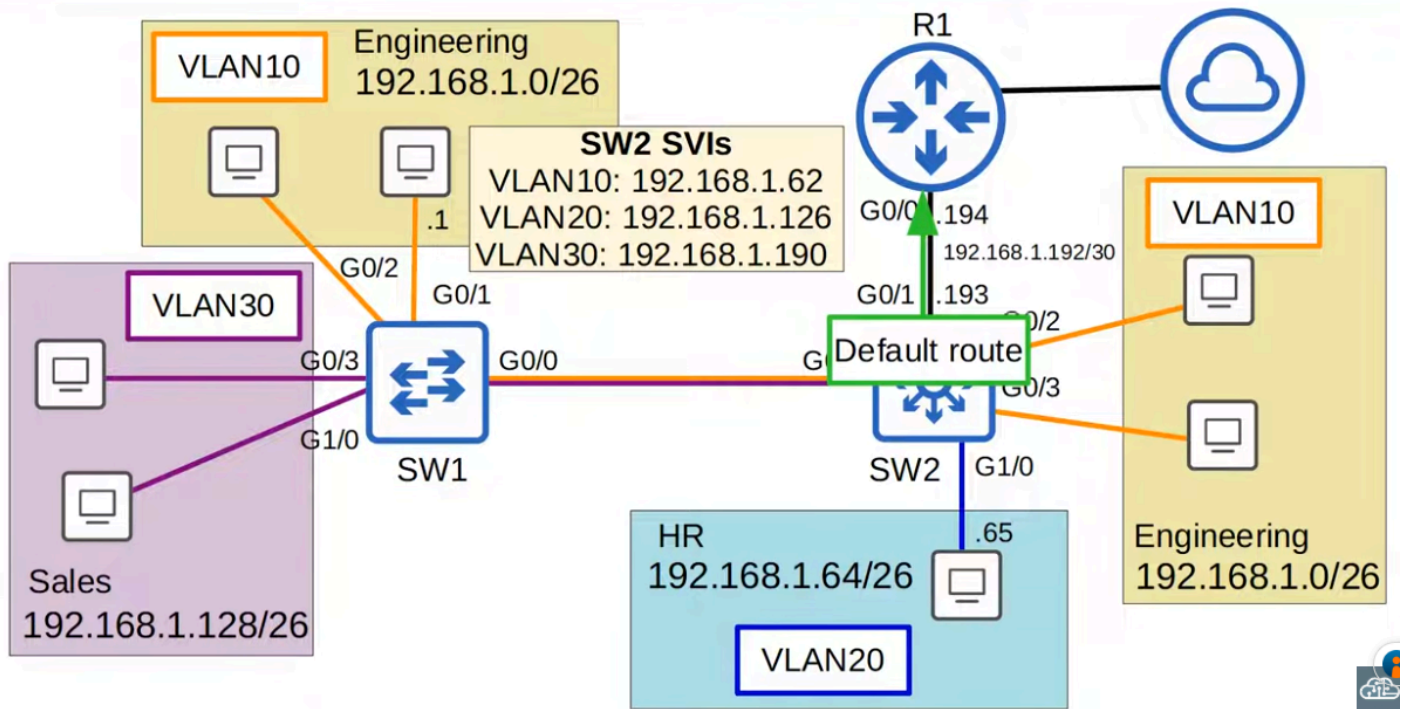
1. The frame **arrives at SW2**
2. The destination is in the **192.168.1.0/26** subnet
3. SW2 now has its **own routing table**, so it looks up the destination
4. SW2 sees that the destination is **connected to its VLAN10 SVI**
5. The traffic is now **routed to VLAN10**
6. If SW2 doesn't have the destination MAC address in its MAC address table, it will **flood** the frame to all VLAN10 interfaces
7. But let's assume it has **already learned the MAC address**
8. SW2 forwards it to **SW1** over its trunk interface, **tagged as VLAN10**
9. SW1 then forwards it to the **destination**

Traffic Destined Outside the LAN

- A **cloud** connected to R1 represents the **Internet**
- Because **SW2 is the default gateway**, any packets destined outside of the local subnet will be sent to **SW2**
- However, the previous router on a stick configurations for the connection between SW2 and R1 will **no longer work**

New Point-to-Point Link Setup

In addition to configuring virtual interfaces (SVIs) on multilayer switches, we can also configure their **physical interfaces** to operate like a router interface, rather than a switchport.



We assign the subnet **192.168.1.192/30** for this point-to-point link:

- **SW2's G0/1** interface: IP address **192.168.1.193**
- **R1's G0/0** interface: IP address **192.168.1.194**

Then we configure a **default route** on SW2 pointing toward R1, so all traffic destined outside of the LAN will be sent to R1.

Configuration of Point-to-Point Link and SVIs

```
R1(config)#no interface g0/0.10
R1(config)#no interface g0/0.20
R1(config)#no interface g0/0.30
R1(config)#default interface g0/0
Interface GigabitEthernet0/0 set to default configuration
R1(config)#do show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       unassigned      YES NVRAM    up          up
GigabitEthernet0/0.10    unassigned      YES manual  deleted     down
GigabitEthernet0/0.20    unassigned      YES manual  deleted     down
GigabitEthernet0/0.30    unassigned      YES manual  deleted     down
GigabitEthernet0/1       unassigned      YES NVRAM    administratively down down
GigabitEthernet0/2       unassigned      YES NVRAM    administratively down down
GigabitEthernet0/3       unassigned      YES NVRAM    administratively down down
R1(config)#
```

Step 1: Remove R1's Router-on-a-Stick Configurations

First, configure the new IP address on R1's G0/0 interface:

Delete each subinterface:

```
no interface g0/0.10
no interface g0/0.20
no interface g0/0.30
```

Reset G0/0 to its default settings:

```
default interface g0/0
```

Check the interfaces:

```
show ip interface brief
```

After this command, notice the status of the subinterfaces — it says **DELETED**. Although we have successfully deleted the subinterfaces, they will remain here with a 'deleted' status **unless we reload the router**. That's no problem though, so we'll just leave them.

Configure the new IP address on G0/0:

```
R1(config)#no interface g0/0.10
R1(config)#no interface g0/0.20
R1(config)#no interface g0/0.30
R1(config)#default interface g0/0
Interface GigabitEthernet0/0 set to default configuration
R1(config)#do show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       unassigned      YES NVRAM    up          up
GigabitEthernet0/0.10    unassigned      YES manual  deleted     down
GigabitEthernet0/0.20    unassigned      YES manual  deleted     down
GigabitEthernet0/0.30    unassigned      YES manual  deleted     down
GigabitEthernet0/1       unassigned      YES NVRAM    administratively down down
GigabitEthernet0/2       unassigned      YES NVRAM    administratively down down
GigabitEthernet0/3       unassigned      YES NVRAM    administratively down down
R1(config)#
```

```
interface g0/0
ip address 192.168.1.194 255.255.255.252
```

This uses a **/30 subnet mask**.

Verify:

```
show ip interface brief
```

You can see that the new IP address has been successfully configured.

Configuring Layer 3 Switching

Important Commands on the Switch

Step 1: Enable IP routing

Also set the g0/1 interface to default configuration because it was configured as a trunk for ROAS because of the previous lab.

```
default interface g0/1  
ip routing
```

This command **enables Layer 3 routing on the switch**. It lets the switch build its own routing table like a router. **If you forget this command, your inter-VLAN routing will not work.**

Step 2: Configure the interface as a routed port

```
interface g0/1  
no switchport
```

The `no switchport` command is what **changes the interface from a Layer 2 switchport to a Layer 3 routed port**. After this, you will be able to assign an IP address to it.

Step 3: Assign an IP address

```
ip address 192.168.1.193 255.255.255.252
```

Step 4: Verify

```
show ip interface brief
```

As you can see, the IP address is assigned to it just like a router interface.

Step 5: Configure the Default Route

Configure a default route pointing to R1:

```
ip route 0.0.0.0 0.0.0.0 192.168.1.194
```

- 192.168.1.194 is R1's IP address (the next hop)

Verify:

```
show ip route
```

You can see that SW2 now has a **routing table** with:

- A **default route** pointing to R1
- **Connected** and **local** routes for the routed interface we configured

Additional Verification Command

```
show interfaces status
```

```
SW2#show interfaces status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/0		connected	trunk	auto	auto	unknown
Gi0/2		connected	10	auto	auto	unknown
Gi0/3		connected	10	auto	auto	unknown
Gi0/1		connected	routed	auto	auto	unknown
Gi1/0		connected	20	auto	auto	unknown
Gi1/1		connected	1	auto	auto	unknown
Gi1/2		connected	1	auto	auto	unknown
Gi1/3		connected	1	auto	auto	unknown
Gi2/0		connected	1	auto	auto	unknown
Gi2/1		connected	1	auto	auto	unknown
Gi2/2		connected	1	auto	auto	unknown
Gi2/3		connected	1	auto	auto	unknown
Gi3/0		connected	1	auto	auto	unknown
Gi3/1		connected	1	auto	auto	unknown
Gi3/2		connected	1	auto	auto	unknown
Gi3/3		connected	1	auto	auto	unknown

Notice that, in the **VLAN column**, instead of a VLAN number, G0/1 displays '**ROUTED**'.

Configuring SVIs on SW2

SVI configuration is **very simple**:

```
interface vlan10
ip address 192.168.1.1 255.255.255.192
no shutdown
```

- SVIs are **shutdown by default**, so remember to use `no shutdown` to enable them
- This is repeated for VLAN20 and VLAN30

Conditions for an SVI to be UP/UP

For a Switch Virtual Interface to become up/up, all of the following must be true:

- **VLAN must exist** on the switch. Creating an SVI does *not* automatically create the VLAN. If the VLAN does not exist, the SVI will remain down/down.

- **At least one access port** in that VLAN is in up/up state, **OR** at least one trunk port that allows the VLAN is up/up.
- **The VLAN itself must not be shutdown.** (Note: this is a different command from shutting the SVI; e.g., `vlan 40` then `shutdown` will disable the VLAN.)
- **The SVI must be enabled** (no `shutdown`).

If a VLAN has no active ports, the SVI will be down/down even if the VLAN exists.

Inter-VLAN Routing Comparison

Method	Description	Pros	Cons
Legacy (separate links)	One physical link per VLAN between router and switch.	Simple, no trunking.	Requires many router interfaces; not scalable.
Router-on-a-Stick	Single trunk link carries all VLANs; router does routing.	Efficient use of interfaces.	Potential bottleneck: all inter-VLAN traffic must go through the router and back.
Multilayer Switching (SVIs)	Switch performs routing internally using SVIs.	Fast (wire-speed routing), scalable, reduces reliance on external router.	Requires Layer 3 switch; configuration must include <code>ip routing</code> and <code>no switchport</code> where needed.

Key Commands Summary

Command	Purpose
<code>encapsulation dot1q <vlan-id> native</code>	Set subinterface as native VLAN on router (Method 1).
<code>ip address <ip> <mask> on physical interface</code>	Configure native VLAN IP directly on router interface (Method 2).
<code>ip routing (global)</code>	Enable Layer 3 routing on a switch.
<code>no switchport (interface)</code>	Change a Layer 2 switchport into a Layer 3 routed port.

<code>interface vlan <id></code>	Create/enter SVI configuration mode.
<code>ip address <ip> <mask> (SVI)</code>	Assign IP to the SVI.
<code>no shutdown (SVI)</code>	Enable the SVI.
<code>ip route 0.0.0.0 0.0.0.0 <next-hop></code>	Configure default route on the switch.
<code>show ip route</code>	Display routing table.
<code>show interfaces status</code>	Show whether a port is switching or routed.

Important Exam Considerations

- **Native VLAN on Router (ROAS):** Two valid methods: (1) subinterface with encapsulation dot1q <vlan> native, or (2) IP on the physical interface.
- **SVI Down/Down:** If an SVI remains down after assigning an IP and `no shutdown`, check whether the VLAN exists and whether there is at least one active port in that VLAN (access or trunk).
- **No Switchport vs. IP Routing:** `no switchport` sets a *single interface* as routed; `ip routing` enables Layer 3 routing globally. Both are needed for a switch to actively route between SVIs and use routed ports.
- **Native VLAN on Trunk:** The default native VLAN is VLAN 1. Changing it with `switchport trunk native vlan <id>` causes traffic in that VLAN to be sent untagged—a common quiz and exam question.